

Rational equations



- 1
- a $x = 0.2$ b $x = 0.45, -0.45$ c $x = 3.5$ d $x = 0.32, -0.32$
e $x = 2.62, 9.38$ f $x = -5$ g $x = 3.29, -5.17$ h $x = 2.53$

- 2
- a $x = -3$
b No it is not, because it makes two of the denominators in the equation equal to 0.

- 3
- a $x = 4$ is not a solution. b $x = 4$ is a valid solution.
c $x = 4$ is not a solution. d $x = 4$ is not a solution.
e $x = 4$ is an extraneous solution. f $x = 4$ is a valid solution.

- 4
- a $x = -\frac{5}{2}$ b $x = -5$ c $x = 28$ d $x = 0$
e $x = 3$ f $x = 7$ g $x = 5, 2$ h $x = -5, -4$
i $x = \frac{3}{2}, \frac{3}{4}$ j $x = 2, -13$

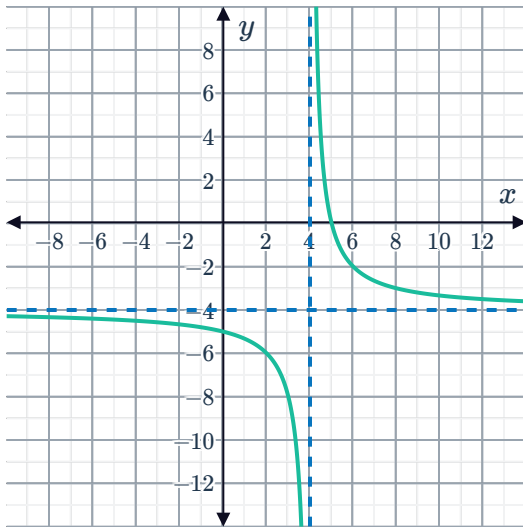
- 5
- a $x = 2$ b $x = 1$ c $x = 4, -4$ d $x = 3, -3$

- 6 $x = 1, -1, -\frac{i}{3}, \frac{i}{3}$

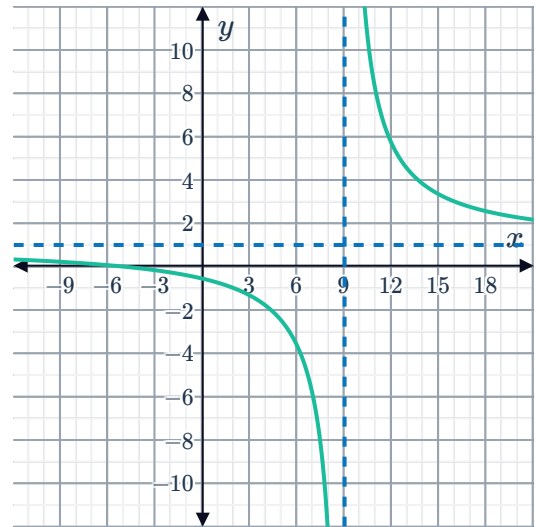
Graphical solutions to rational equations

- 7
- a i 1 ii 1 iii 1 iv 1 b i 2 ii 1 iii 0 iv 1
c i 1 ii 1 iii 1 iv 2 d i 2 ii 1 iii 1 iv 2

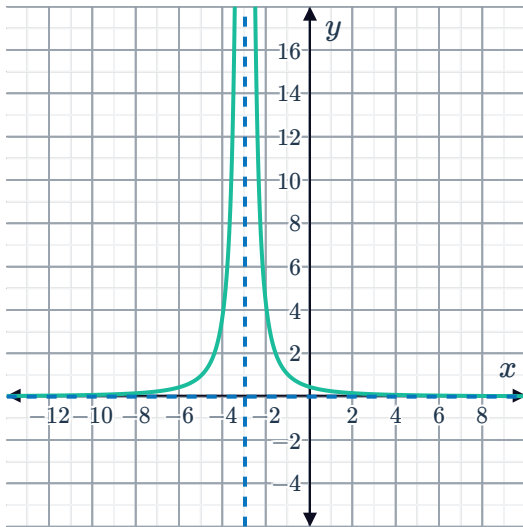
a



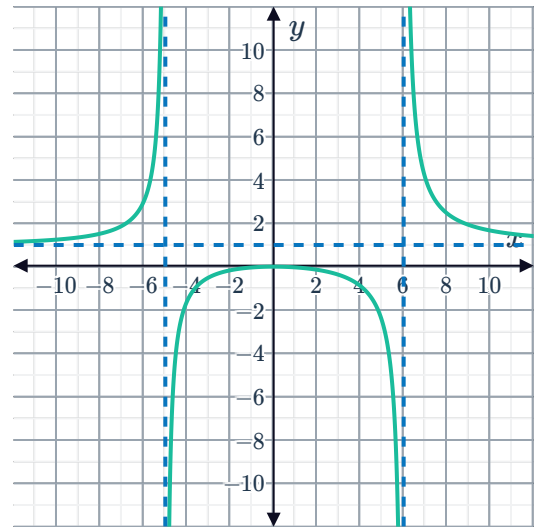
b



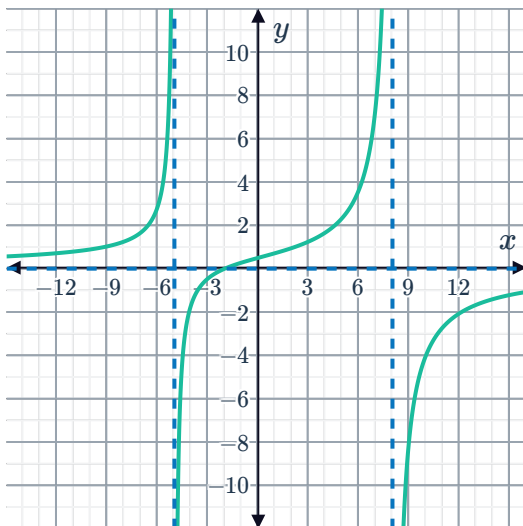
c



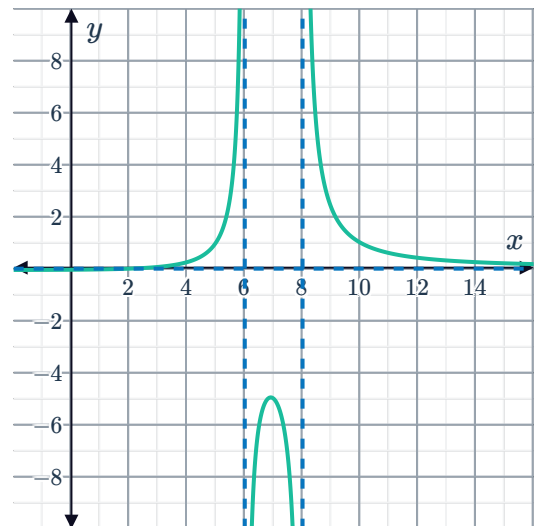
d



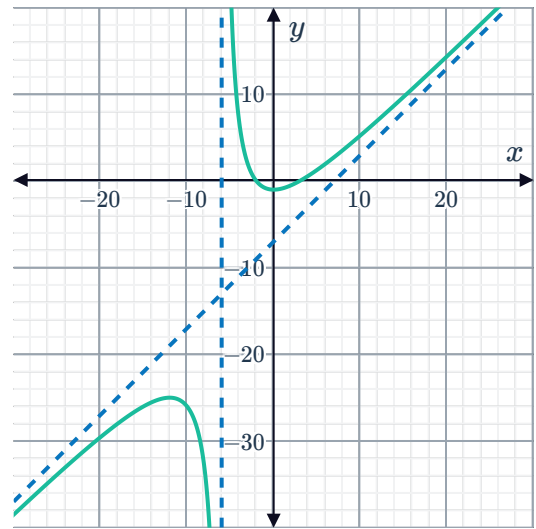
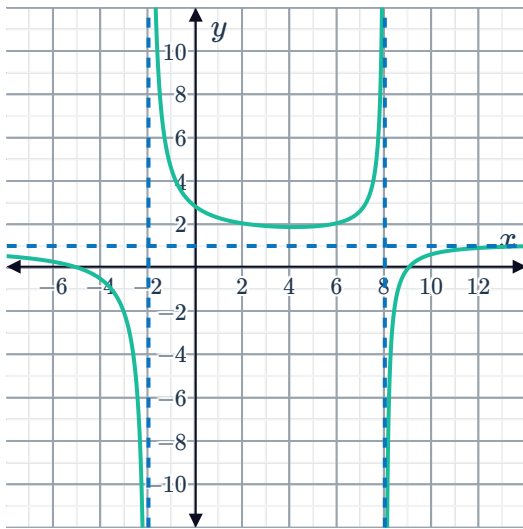
e



f

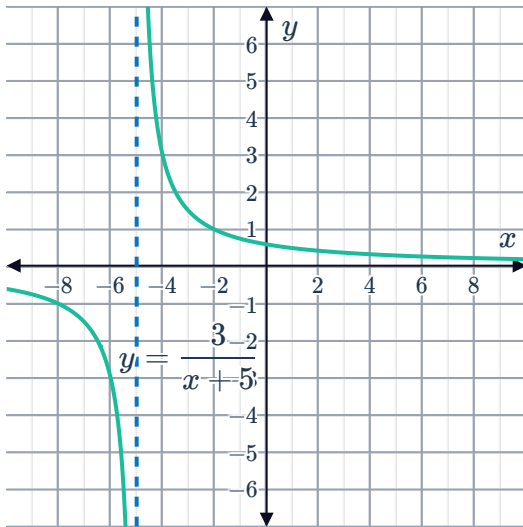


g

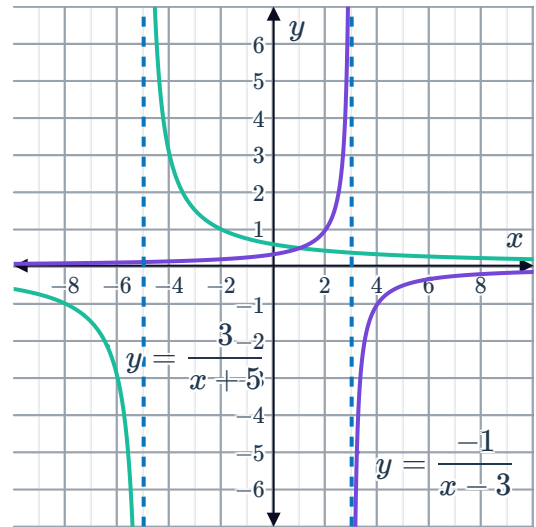


9

a



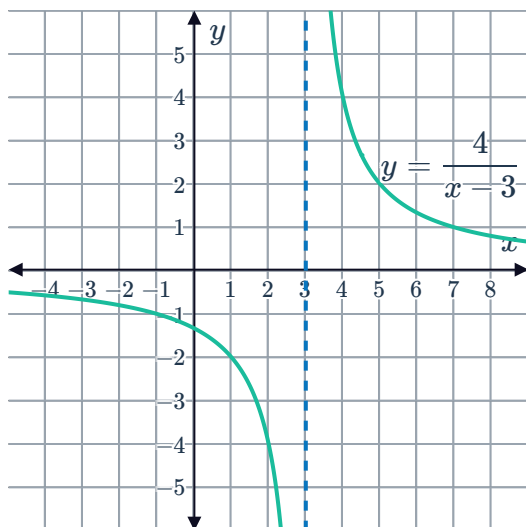
b



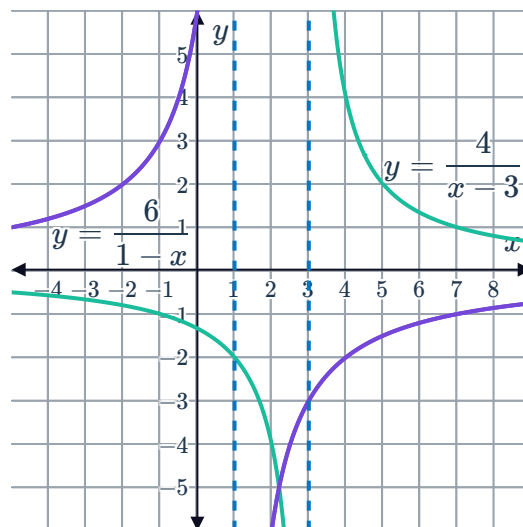
c $x = 1$

10

a



b

c $x = 2.2$